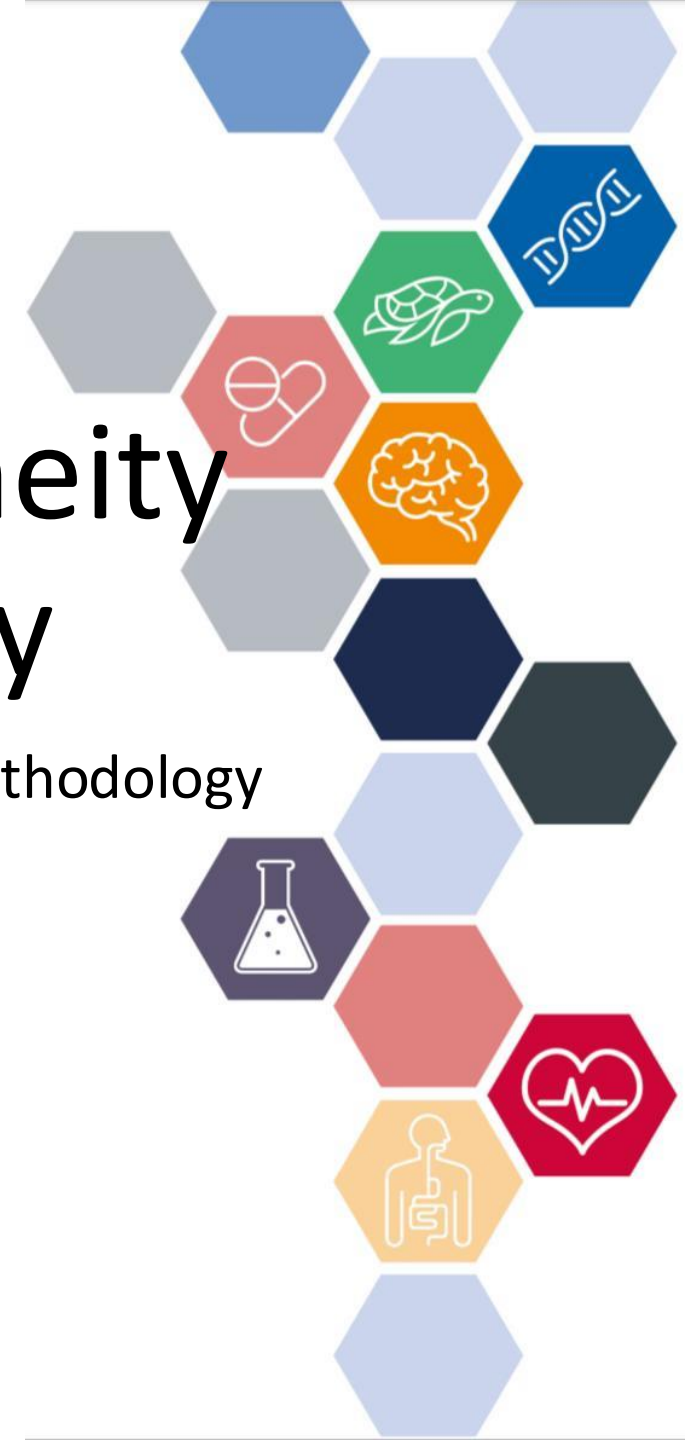


Mapping the pH Heterogeneity of the Secretory Pathway

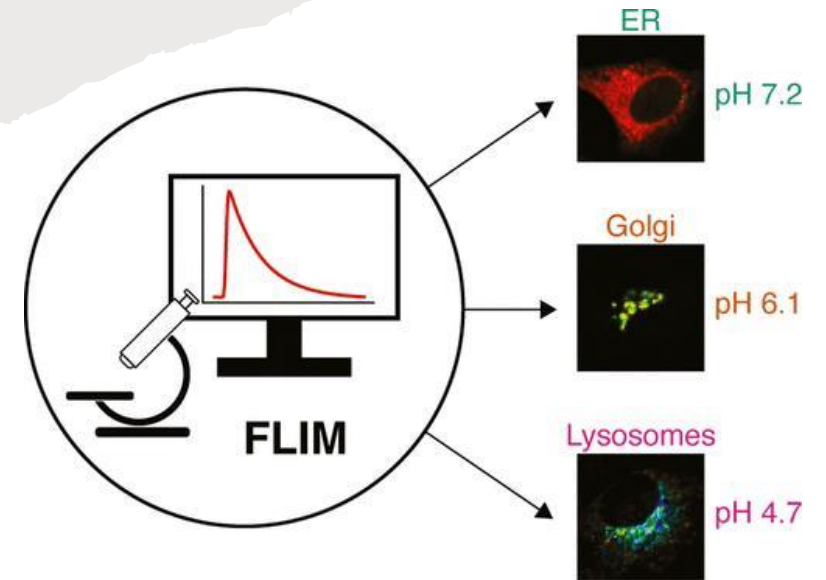
Development and Critical Evaluation of a Guided Unmixing Methodology

Treananan Gerald Yuan



Intracellular pH Gradients are Critical for Cell Function

- pH regulates protein folding, enzyme activity, trafficking
- Disruptions linked to Congenital Disorders of Glycosylation (CDG)
- Lysosomal storage diseases result from pH imbalance



Organelle	pH
Endoplasmic Reticulum (ER)	~7.2
Golgi Apparatus	~6.0 – 6.7
Secretory Vesicles	~5.2
Lysosomes	~4.7

FLIM Data Requires Unmixing

What was measured

- A single pixel's decay curve
- It's a superposition of multiple components:
 - pH sensor in different protonation states
 - Auto-fluorescence (NADH, FAD)
 - Other fluorophores (Probes)

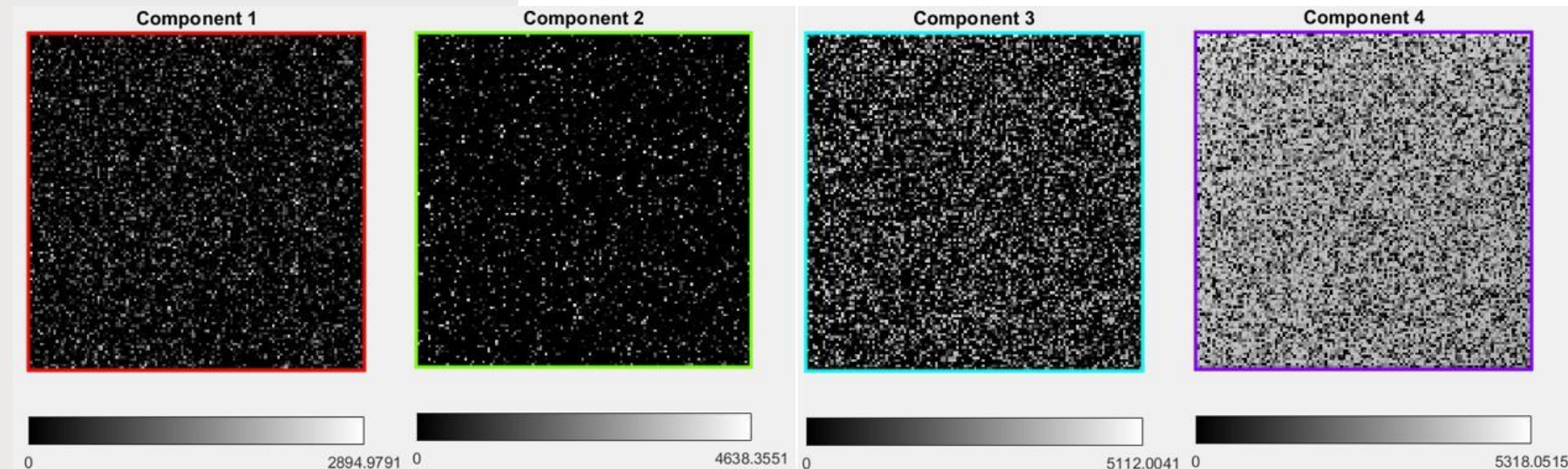
What we want

- Four clean component images separated

Non-negative Matrix Factorisation (NMF) solves this

Equation: $D \approx S \times T$

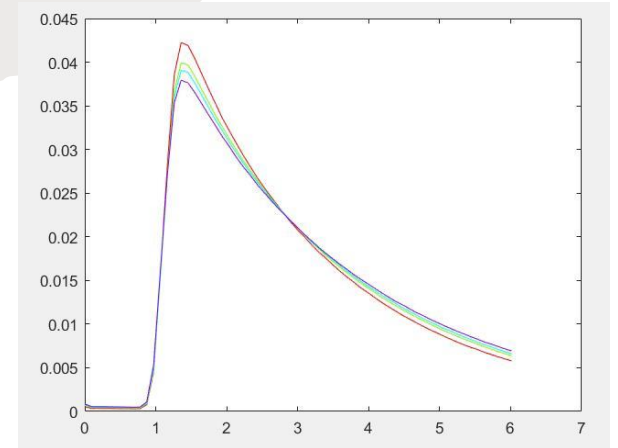
D (Data) = S (Spatial Maps) x T (Decay Curves)



Project Aims and Hypothesis

1. Calibrate

- Generate reference decay curves from recombinant RpHluorin2
- pH values: 4.5, 5.5, 6.0, 7.5 (Good peak separation)

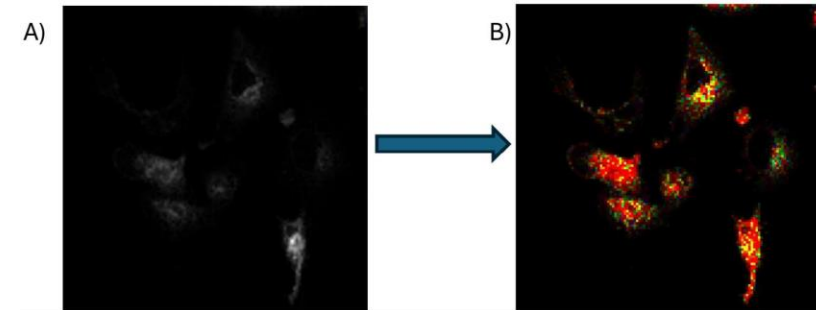


2. Implement Guided NMF

- Use reference curves as fixed components
- Analyse cellular FLIM data from Linders et al. (2022)

3. Generate pH Maps

- Weighted average: $\text{pH}_{\text{estimated}} = (\sum_i \text{pH}_i \times C_i) / (\sum_i C_i)$
- i is the component index, pH_i is the nominal pH of each reference component (4.5, 5.5, 6.0, 7.5), and C_i is the abundance (concentration) of that component in the pixel



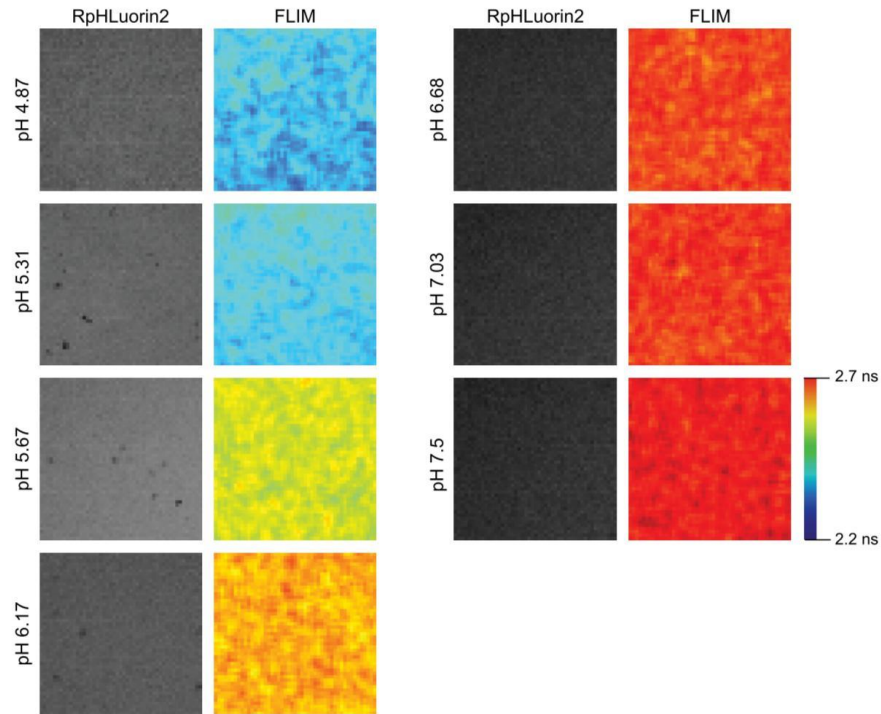
4. Critically Evaluate

- Assess accuracy, limitations, SNR requirements
- Propose future improvements

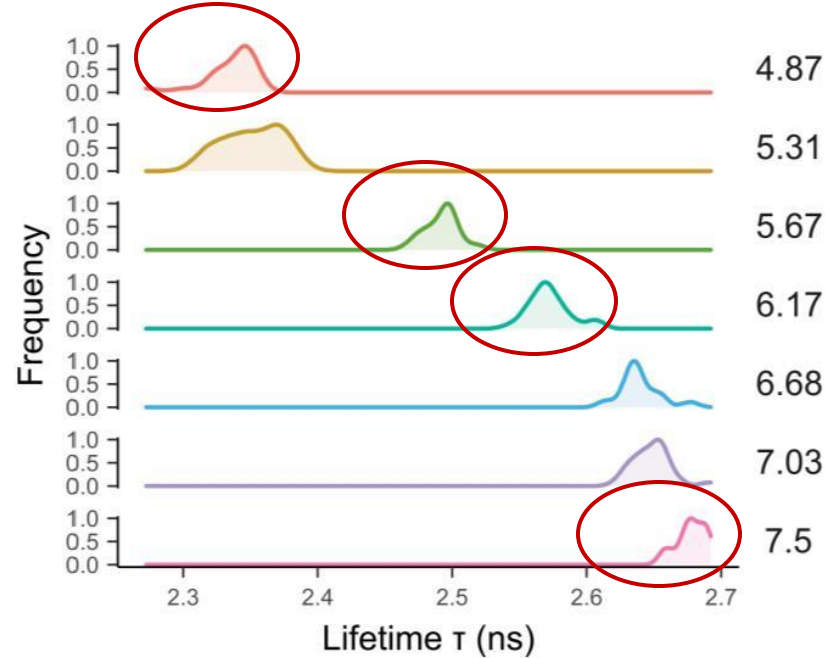
Hypothesis: *in vitro* decay curves can guide accurate pH mapping in cellular data

Generating Reference Decay Curves

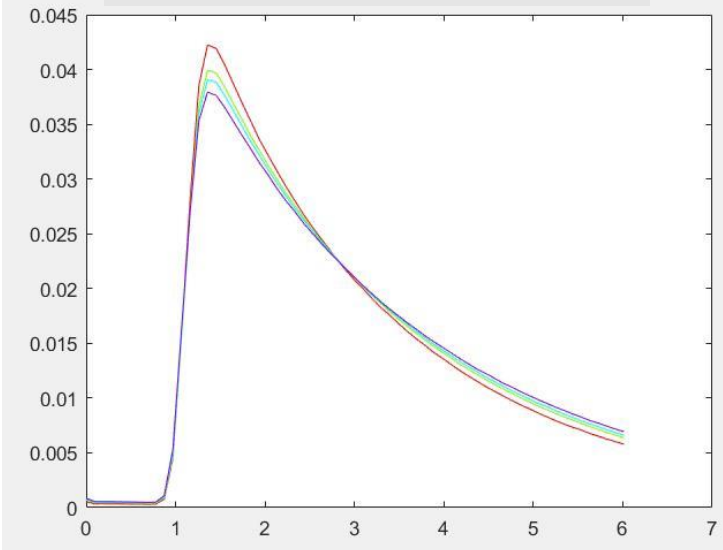
FLIM data acquisition



Raw decay curves for pH 4.5 to 7.5



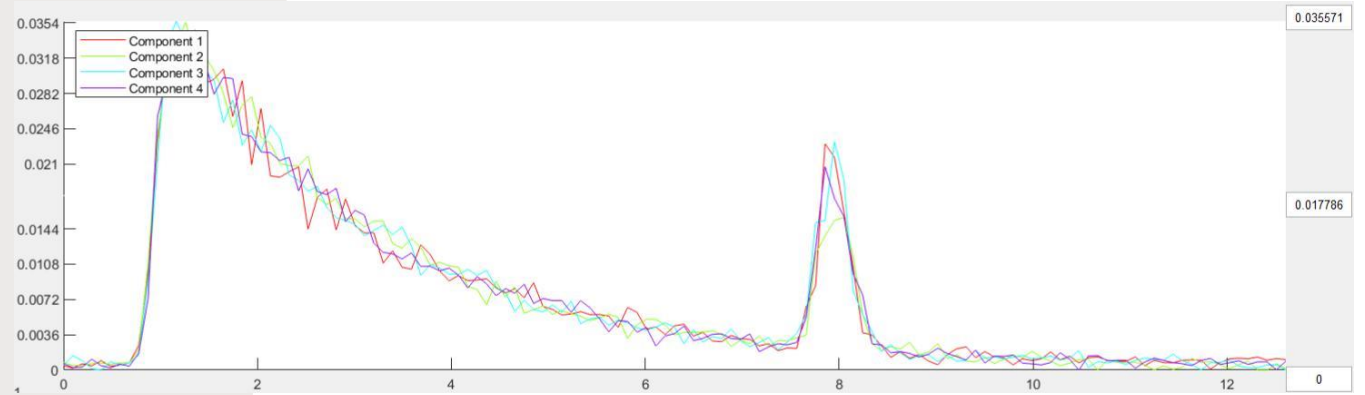
Selected reference curves (4.5, 5.5, 6.0, 7.5) – Normalised



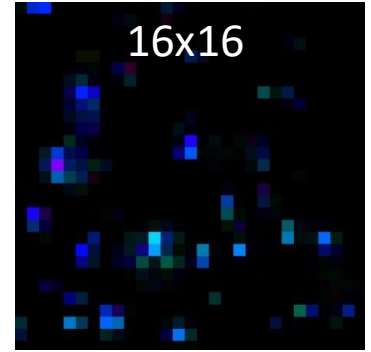
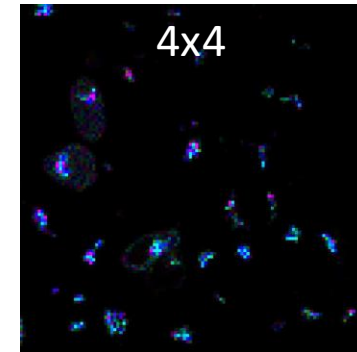
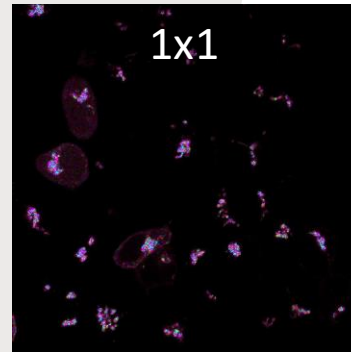
Four curves compiled into ASCII text file as fixed reference matrix

Pre-processing Optimisation

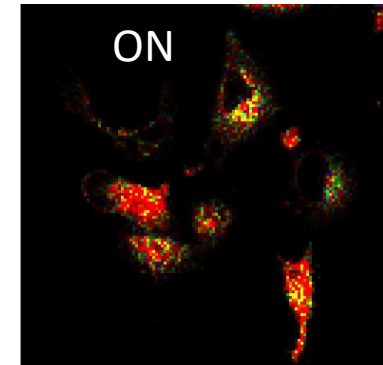
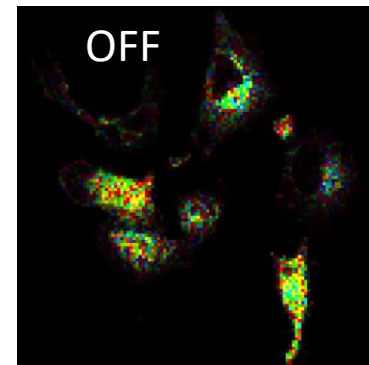
Spectral Cropping



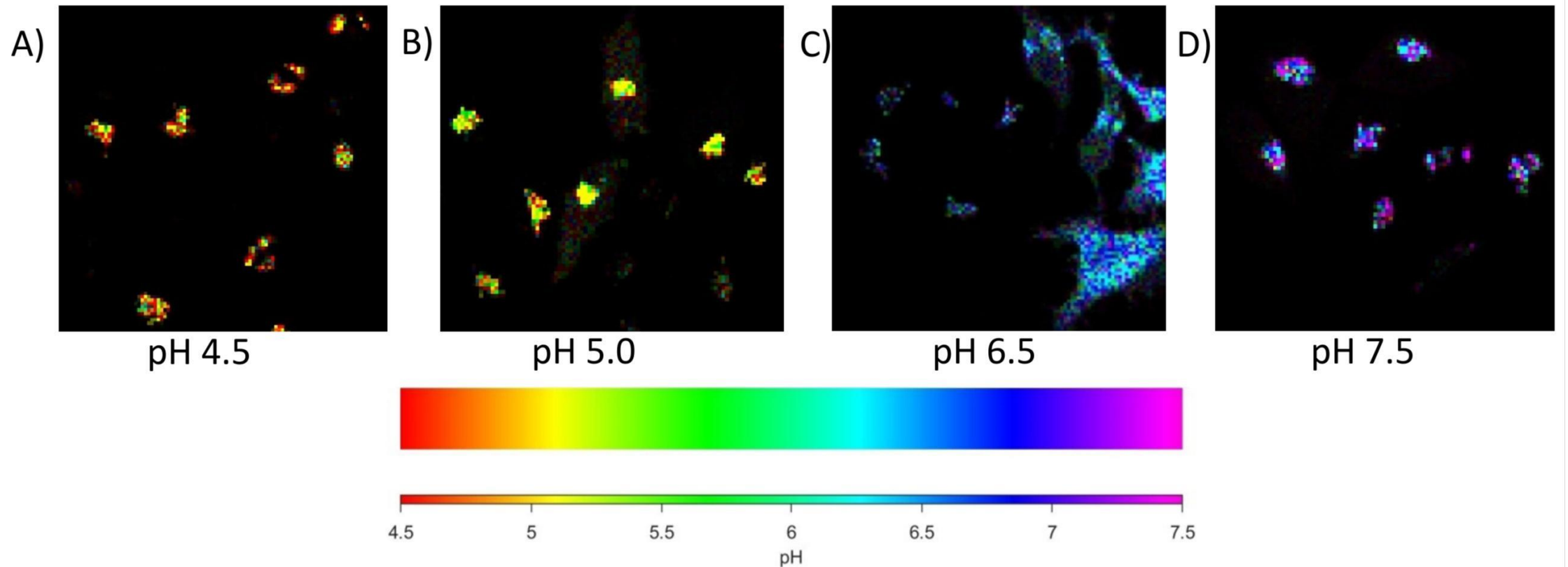
Spatial Binning Optimisation



Partial Noise Whitening

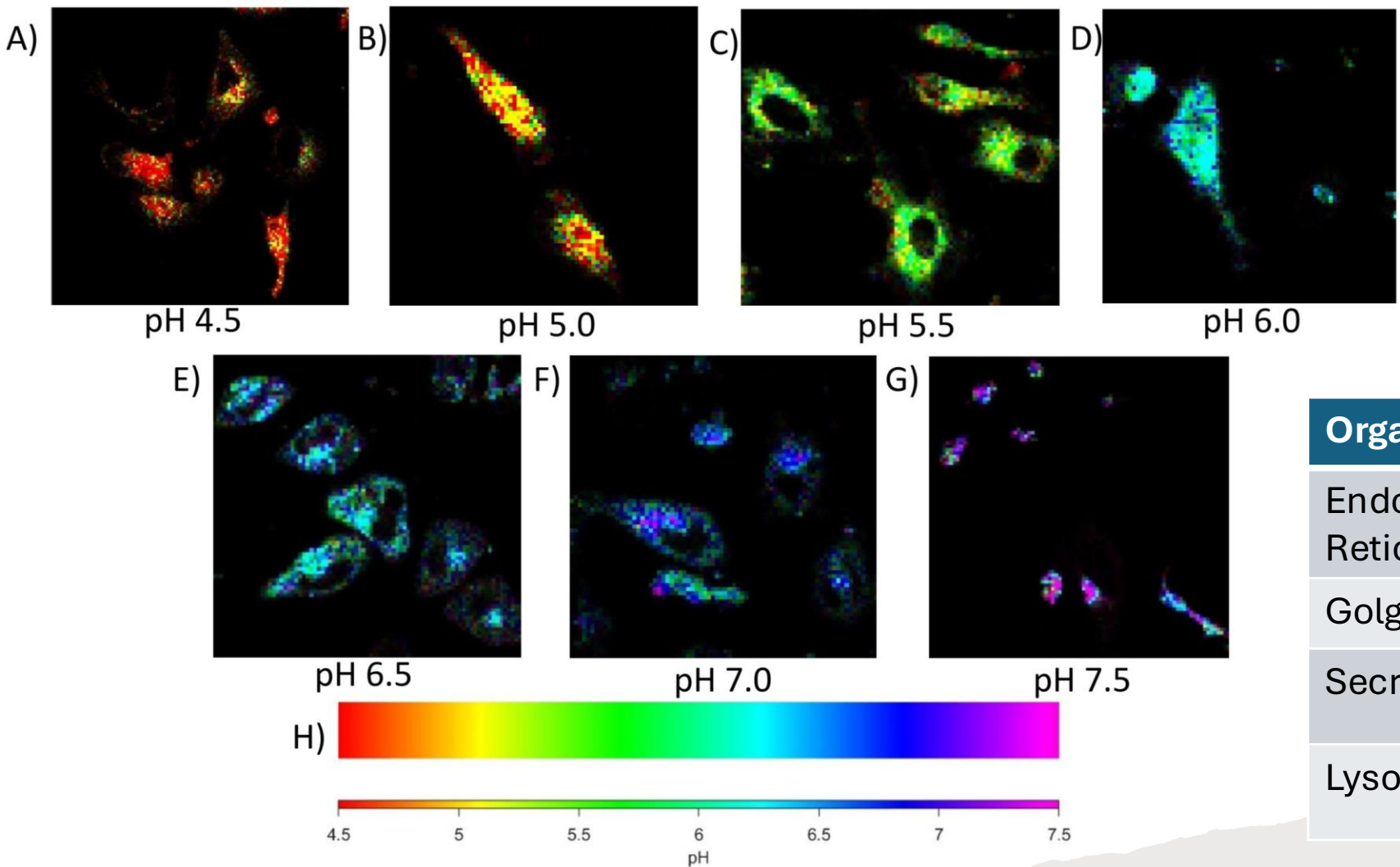


Validation: Recombinant Calibration Samples



Algorithm correctly assigns pH based on reference decays

Cellular Results: pH Maps Match Expected Values



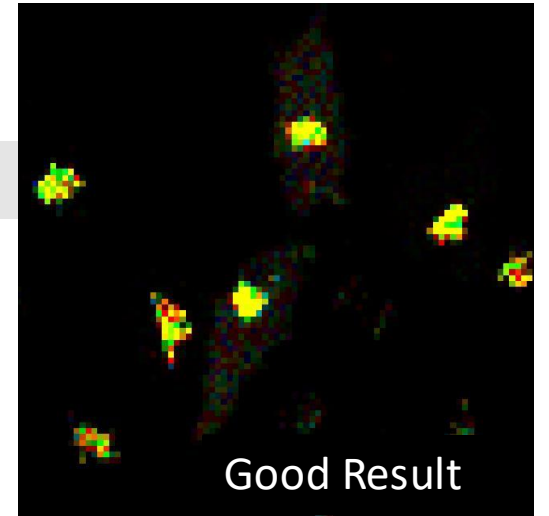
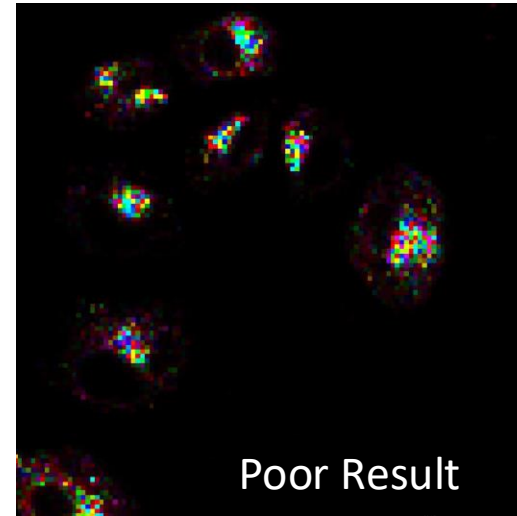
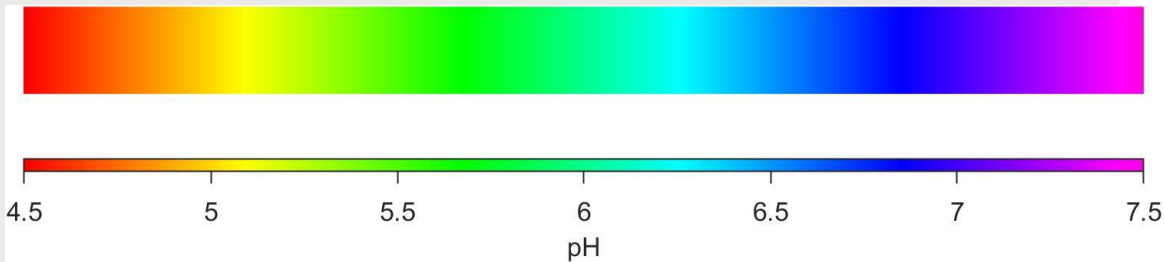
Maps show expected pH for each organelle

Organelle	pH
Endoplasmic Reticulum (ER)	~7.2
Golgi Apparatus	~6.0 – 6.7
Secretory Vesicles	~5.2
Lysosomes	~4.7

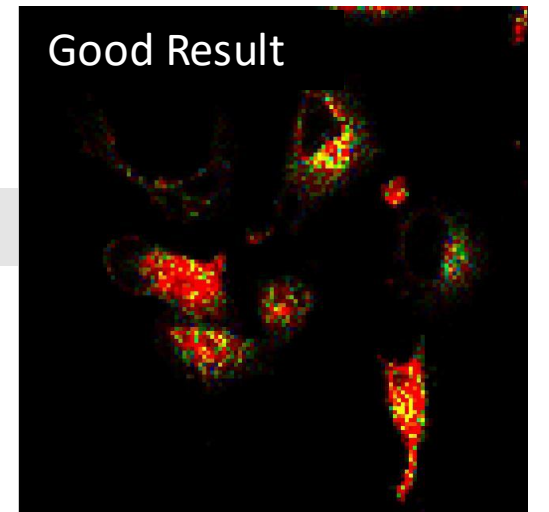
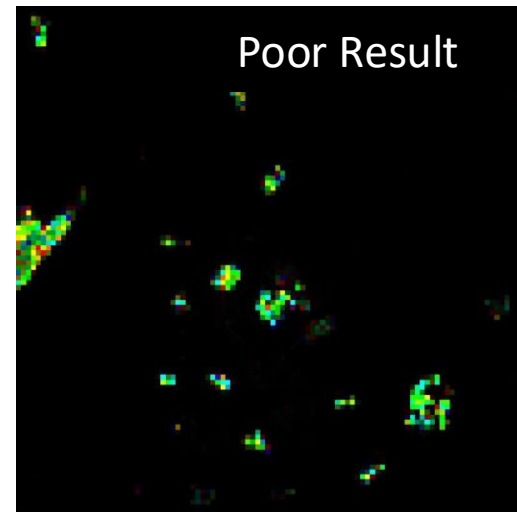
Not All Datasets Succeed: Data Quality Matters

Possible Explanations:

- Different acquisition days
- Lower excitation power: Fewer protons
- pH instability
- Optical misalignment



Same analysis using 4x4 binning, same references, same guided NMF



Identified Limitations

Model Rigidity

- Fixed *in vitro* references may not match cellular environment
- Crowding, viscosity alter photophysics

SNR Dependence

- 4x4 binning essential
- Trade-off: Resolution vs Reliability
- Precision related to photon budget

Reference Mismatch

- Some offsets observed
- Cellular environment $\neq$ Cuvette
- Supervised approach vulnerable to divergence

Photon budget is fundamental: To double resolution, we need to increase the photons four times

Future Directions: From Supervised to Unsupervised

uFLIM

- Solves for both S and T simultaneously ($D \approx S \times T$)
- No references needed
- Discovers true component dynamics
- Separates autofluorescence automatically

Hybrid Approach

- Guided analysis with pH 4.5/7.5 only
- uFLIM refines, discovers intermediate states

Multi-parameter Imaging

- Co-express pH sensor + FRET biosensor
- Single excitation, simultaneous readouts
- pH + calcium + activity maps

Conclusion

- Guided NMF successfully maps pH
- Maps match expected organelle pH
- Limitations identified:
 - SNR dependence
 - Model rigidity
 - Data quality

While guided FLIM unmixing works, unsupervised methods are the future.

Unsupervised uFLIM: More robust, quantitative, comprehensive cellular mapping